COMP10020 · INTERNET TECHNOLOGIES · GROUP H

Umami: Counting Visitors Without Cookies

Open-source, privacy-focused web analytics



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01 THE PROBLEM

The analytics dilemma



Sites need data

- How many visitors, from where
- Which pages attract attention



But cookies have a cost

- 60%+ of popular EU sites showed consent notices (Utz et al., 2019)
- Hiding first-page ‘reject’ raised consent by 22–23 percentage points (Nouwens et al., 2020)

“

Can we get useful analytics without relying on cookies?

02 MEET THE TOOL

What is Umami?

An open-source platform for web and product analytics



Open source

Public source code for inspection and learning



Cookie-free tracker

Records website activity without analytics cookies



Hosting options

Managed Umami Cloud or a self-hosted instance

03 MEET THE TOOL

What Umami measures

Page activity, visitor context and selected actions



Traffic overview

Pageviews and visitor activity over time



Visitor context

Referrers, devices, browsers and approximate locations



Custom events

Record selected actions and add properties for context

04 MEET THE TOOL

UTM tracking and sharing

UTM reports

- Label links by source and campaign
- Compare views from email or social posts
- Same page, different tagged links

Share URLs

- Publish selected analytics views
- View reports without an admin account
- Choose shared information carefully

Next: how does Umami count visitors without cookies?

05 ARCHITECTURE

How the parts connect



Two filters before saving: ① **bot check** (isbot library) ② **IP blocklist** (e.g. your own office)

No cookie at any step. So what is inside that request?

06 THE DATA

What Umami sees, and what it doesn't



① Sent by script.js

- Page URL and title
- Referring site
- Screen size
- Browser language



② Added by the server

- Browser and operating system (from user agent)
- Device type
- Country, region, city (from IP)



③ Never stored

- The IP address itself
- Cookies of any kind
- Names or emails (unless the owner sends them)

No cookie anywhere. So how does it know two page views are the same visitor?

07 HASHING

Hashing: a fingerprint, not a cookie

Same input → same code

Code → can't go back to the input

Oct · morning	site + IP + browser + October salt	a3f9c2…
Oct · afternoon	site + IP + browser + October salt	a3f9c2…
November	site + IP + browser + November salt	7d10e8…

Codes match → 1 visitor, nothing stored on the device

Salt changes monthly → new code, no link to October

Codes are illustrative; real IDs are UUIDs built from the hash.

08 UNDER THE HOOD

The exact formulas

```
Session ID  =  hash( website ID + IP + user agent + monthly salt )
```

```
Visit ID  =  hash( session ID + hourly salt )
```

Checked in Umami's source: `src/app/api/send/route.ts`

1. **IP** used in the hash, never saved
2. **Monthly salt** no tracking across months
3. **"Visitors"** = sessions, not identified people
4. **30 min idle** = new visit (bounce rate, duration)

09 TRADE-OFFS

What this design costs

Benefits

- No identity in the database
- No cookie ID, no stored IP
- Small script, quick setup

Costs

- Wi-Fi → 4G = counted as a new visitor
- Shared Wi-Fi + same browser = two people may merge
- Can't follow one user over months
- Hashed ID is pseudonymous, still personal data (EDPB, 2025)
- Self-hosted: you set data retention

Precision ↔ privacy: no perfect answer, only a choice.

10 DEMONSTRATION: THE WEBSITE

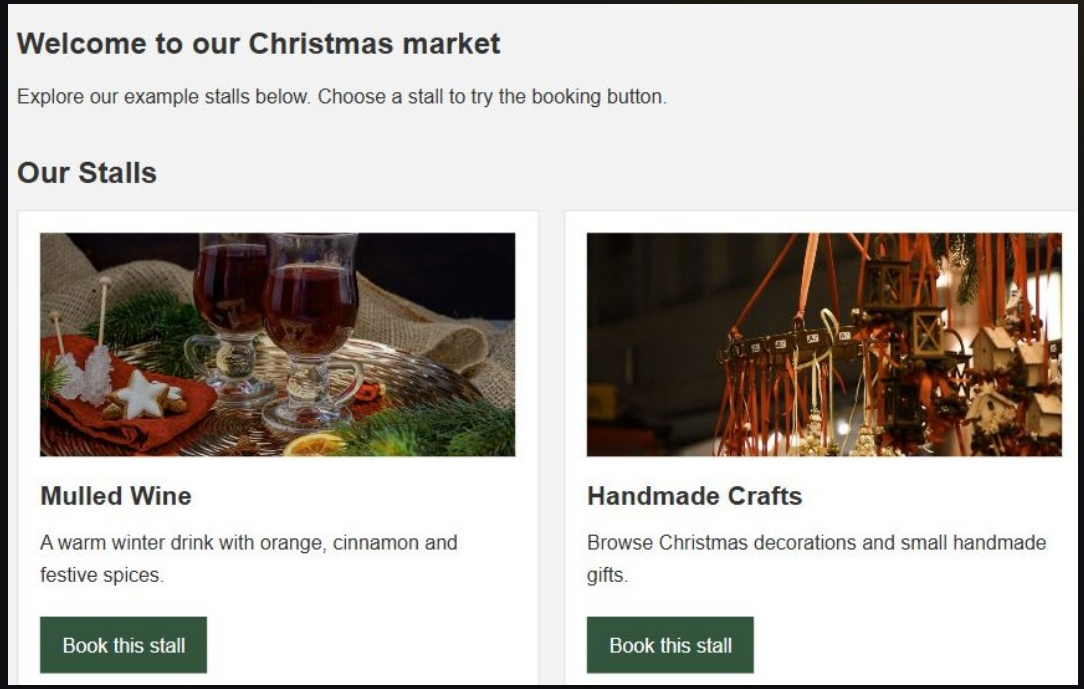
Our demo website

Paisley Christmas Market

A local demonstration with two stall buttons

Which stall button receives more clicks?

http://localhost:8080/



Connecting to Umami Cloud

Our configuration: umami-config.js

umami-config.js

```
// Umami Cloud tracking for the Paisley Christmas Market p
window.PAISLEY_UMAMI = {
  scriptUrl: 'https://cloud.umami.is/script.js',
  websiteId: '230853b1-91d3-4f91-9553-0d6a20e02482',
};
```

scriptUrl

Loads the Umami Cloud tracker

websiteId

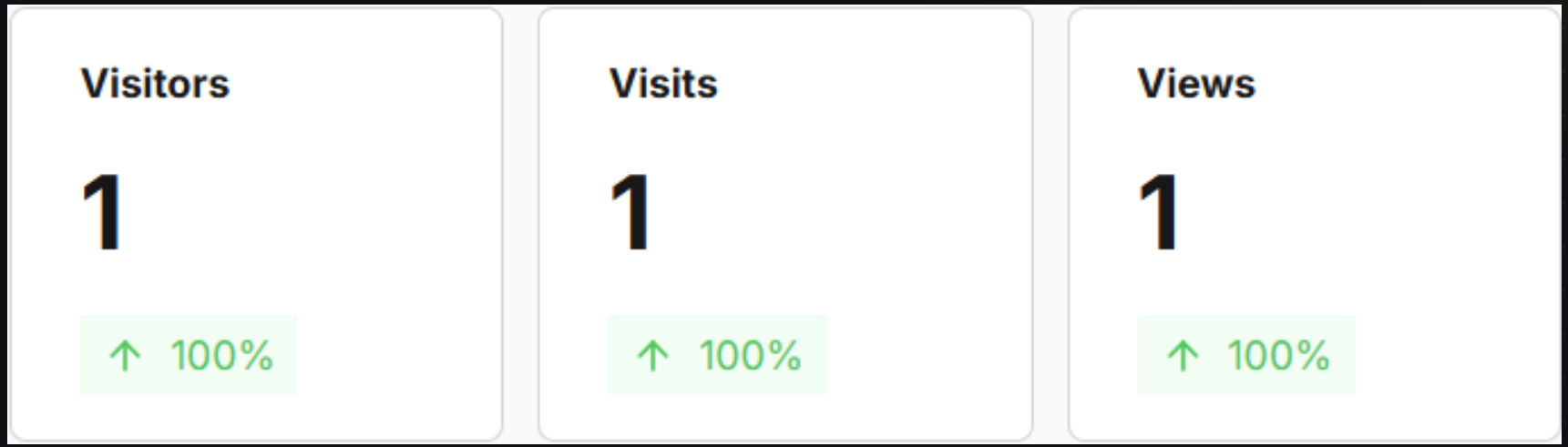
Links this website to its dashboard

index.html loads this configuration before app.js adds the tracker.

12 DEMO 1: VERIFYING PAGEVIEWS

Checking the first page visit

Our first visit appeared in the Overview dashboard



The tracker is sending page visits to Umami.

13 DEMO 2: ADDING AN EVENT

Tracking a button click

Our HTML button in index.html

umami-config.js | index.html

index.html > ...

```
37  <!-- Umami event: Book stall. Property: stall = mulled-wine. -->
38  <button type="button"
39    data-umami-event="Book stall"
40    data-umami-event-stall="mulled-wine"
41    data-stall-name="Mulled Wine">Book this stall</button>
```

Event name

Book stall

Event property

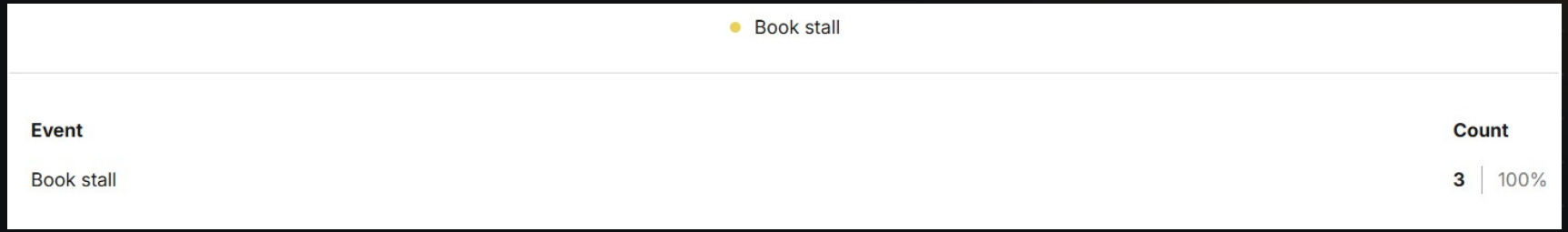
stall = mulled-wine / crafts

A button click records an action, not a confirmed booking.

14 DEMO 2: CHECKING THE EVENT

Checking the recorded clicks

Test snapshot: 3 clicks on Book stall



1 Click one stall button on our website.
2 Switch to Umami Events and refresh.
3 Check the updated count.

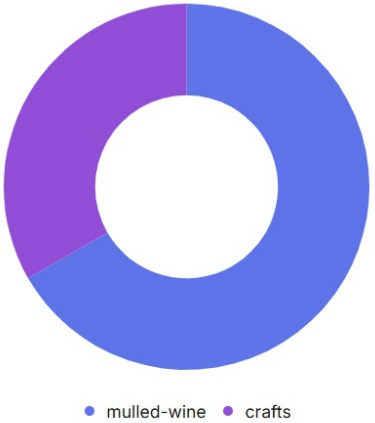
Further clicks will change the count shown during the live demonstration.

15 DEMO 3: COMPARING STALLS

Reading event properties

Events / Properties / stall

stall	Count	
mulled-wine	2	67%
crafts	1	33%



Mulled Wine: 2 clicks (67%) Handmade Crafts: 1 click (33%)

This small test confirms tracking. It does not establish customer demand.

16 USE CASE

Paisley Christmas Market



Which stalls get the most interest ?

Top pages + Book stall clicks, grouped by stalls

Where do visitors come from?

Referrers: social media, search or direct

Are visitors mainly on phone or desktop?

Device breakdown for a mobile-first design

Where do bookings drop off?

Funnel: stall page → booking form → booking confirmed

Requires tracking the booking confirmation. Our demo records button clicks, not completed bookings.

17 COMPARISON

How it compares

	Umami	Google Analytics	Matomo
Open source	Yes	No	Yes
Self-hosting	Yes	No	Yes
Cookies in the default tracker	No	Use first-party cookies; Consent settings affect behaviour	Use first-party cookies by default; Can be disabled
Price	Free self-hosted; Cloud plans	Free; GA360 paid	Free self-hosted; Cloud paid
Feature depth	Focused	Very deep, ads integration	Deep
Hosting and maintenance	Self-hosted or managed Cloud	Google-hosted	Self-hosted or managed Cloud

(Google Analytics, n.d.; Matomo, n.d.; Umami Software, n.d.-a)

18 VERDICT

When (not) to use Umami



Use it when...

- You need basic traffic and event reports without cookies
- You want to self-host and control where analytics data is stored



Avoid it when...

- You need deep ad-platform integration
- You need identity-level tracking across months

Useful analytics without cookies? Yes — if we accept estimated visitor counts rather than identified individuals.

19 SOURCES

References

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Thank you

Questions?

